

UNIT 2 ASSESSMENT OF LEARNING: EXPLORING DERIVATIVES – DAY 1

Instructions

- Non-graphing calculators may be used but not shared.
- Notebooks may not be used. You MUST use concepts covered in this unit/course. Show all steps for full marks.
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K & U	Thinking	Comm.
/20	/5	/2

Knowledge and Understanding - [20 Marks]

Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided.
[1 Mark Each – 5 Marks Total]

- If $y = x^3 f\left(\frac{1}{x^2}\right)$, then $\frac{dy}{dx}$ is C
 - $x^3 f'\left(\frac{1}{x^2}\right)$
 - $3x^2 f\left(\frac{1}{x^2}\right) + x^3 f'\left(\frac{1}{x^2}\right)$
 - $3x^2 f\left(\frac{1}{x^2}\right) - 2f'\left(\frac{1}{x^2}\right)$
 - $3x^2 f'\left(\frac{1}{x^2}\right)$
- If $y = 2x^4 - 3x^3 - 6x + 23$, then $\left.\frac{d^2y}{dx^2}\right|_{x=-1}$ equals A
 - 42
 - 25
 - 30
 - 78
- If $s(t) = \frac{5}{2}t^3 - t^2 + 4t$ is the displacement, in metres, of a particle at time, t seconds, then initial velocity is equal to B
 - 1 m/s
 - 4 m/s
 - 0 m/s
 - $\frac{5}{2}$ m/s
- If $f(x) = 3\sqrt{x} - 2$, then the slope of the **normal** line at $x = 16$ is equal to D
 - $-\frac{3}{8}$
 - $\frac{3}{8}$
 - $\frac{8}{3}$
 - $-\frac{8}{3}$
- Which of the following is/are **true**? A
 - If $g(x) = 4f(4x)$, then $g'(x) = 16f'(4x)$.
 - If $f(x) = (\pi + 1)^5$, then $f'(x) = 5(\pi + 1)^4$.
 - If $f(x)$ is a degree 10 polynomial function then $f'''(x)$ is a degree eight polynomial function.
 - If $f(x) = g(x)h(x)t(x)$, then $f'(x) = g'(x) + h'(x) + t'(x)$.
 - I only
 - II only
 - I, II and IV only
 - I and IV only
- Differentiate the following with respect to x . **Do Not** simplify your final answers. If applicable, leave answers with **negative exponents**. [6 Marks]

a. $y = \frac{7x^5 - x^{\sqrt{8}} + 3\sqrt[4]{x^3}}{x}$

[3]

b. $y = \frac{1}{(x+5)^2} + \sqrt[3]{2 - (5x-9)^{11}} - \frac{7}{3}$

[3]

$y = 7x^4 - x^{\sqrt{8}-1} + 3x^{-\frac{1}{4}}$

$y' = -2(x+5)^{-3} + \frac{1}{3}(2 - (5x-9)^{11})^{-\frac{2}{3}}(-11(5x-9)^{10})$

7. Differentiate the following with respect to x . Completely simplify your answers. If applicable, write all answers with positive exponents. [9 Marks]

a. $f(x) = (8x+3)\sqrt{4x^2+3x}$ [4]

$$\begin{aligned} f'(x) &= 3(8x+3)^2 \cdot 8\sqrt{4x^2+3x} + (8x+3)^3 \cdot \frac{1}{2} (4x^2+3x)^{-\frac{1}{2}} (8x+3) \\ &= (8x+3)^2 \left(24\sqrt{4x^2+3x} + \frac{(8x+3)^2}{2\sqrt{4x^2+3x}} \right) \\ &= (8x+3)^2 \left(\frac{48(4x^2+3x) + (8x+3)^2}{2\sqrt{4x^2+3x}} \right) \\ &= (8x+3)^2 (256x^2 + 192x + 9) \end{aligned}$$

b. $h(x) = \frac{2\sqrt{4x^2+3x}}{(7-4x)^5(2x^2+5)^4}$ [5]

$$\begin{aligned} h'(x) &= \frac{5(7-4x)^4(-4)(2x^2+5)^4 - (7-4x)^5 \cdot 4(2x^2+5)^3 \cdot (4x)}{(2x^2+5)^8} \\ &= \frac{-4(7-4x)^4(2x^2+5)^3 [5(2x^2+5) + 4(7-4x)(x)]}{(2x^2+5)^8} \\ &= \frac{-4(7-4x)^4(2x^2+5)^3 [-6x^2 + 28x + 25]}{(2x^2+5)^8} \\ &= \frac{4(7-4x)^4(6x^2 - 28x - 25)}{(2x^2+5)^5} \end{aligned}$$

Thinking - [5 Marks]

1. The tangent to the curve $f(x) = (6x)\sqrt{1+4x}$ at the point $(2, 36)$ meets the x -axis at A and the y -axis at B. Determine the exact area of the triangle OAB where O is the origin. [5 Marks]

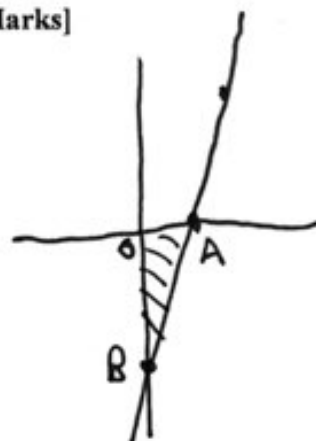
$$f'(x) = 6\sqrt{1+4x} + \frac{6x \cdot 4}{2\sqrt{1+4x}}$$

$$\begin{aligned} f'(2) &= 6(3) + \frac{48}{6} \\ &= 18 + 8 \\ &= 26 \end{aligned}$$

$$y - y_1 = m(x - x_1)$$

$$y - 36 = 26(x - 2)$$

y-int: x -int:



$$\text{Area} = \frac{1}{2} b \cdot h$$

$$\text{Area} = \frac{1}{2} (8)(16)$$

UNIT 2 ASSESSMENT OF LEARNING: EXPLORING DERIVATIVES – DAY 2

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Application	Thinking	Comm.
/20	/5	/2

Application - [20 Marks]

1. Water is flowing out of a tank. The volume, V , in litres remaining after t minutes is given by $V(t) = 1000(20 - t^2)$.

- a. How fast is the water leaving the tank when it is half full? Round your answer to 2 decimal places. [2 Marks]

$$10000 = 1000(20 - t^2) \quad V'(t) = -2000t$$

$$\frac{1}{2} = 20 - t^2 \quad V'(\sqrt{10}) = -2000\sqrt{10} \text{ L/min.}$$

$$t^2 = 10 \quad V'(\sqrt{10}) = -6324.56 \text{ L/min}$$

$$t = \sqrt{10}$$

- b. How much water is in the tank when it is flowing out at a rate of 4,000 litres per minute? Round your answer to 2 decimal places. [3 Marks]

$$V'(t) = -4000 \quad V(2) = 1000(20 - 4)$$

$$-4000 = -2000t \quad V(2) = 16,000 \text{ L}$$

$$t = 2$$

2. Determine the value of a if the graph of $f(x) = \frac{1}{ax+2}$ has the tangent line $3x + 4y - 2 = 0$ at the point

$(0, \frac{1}{2})$. [4 Marks]

$$y = -\frac{3}{4}x + \frac{1}{2}$$

$$f(x) = \frac{-a}{(ax+2)^2}$$

$$f'(0) = \frac{-a}{4} = -\frac{3}{4}$$

$$\therefore a = 3$$

3. Determine the equation (point-slope form) of the tangent to the graph of $y = -3x\sqrt{x} + 2$ at the point where $x = 4$. [3 Marks]

$$y' = -\frac{9}{2}x^{\frac{1}{2}}$$

$$y' = -\frac{9}{2}(4)^{\frac{1}{2}}$$

$$y' = -9$$

$$y - y_1 = m(x - x_1)$$

$$y + 22 = -9(x - 4)$$

$$\rightarrow = -3x^{\frac{3}{2}} + 2$$

4. Using Leibniz's Notation, determine $\frac{dy}{dx}\bigg|_{x=5}$ given $y = 2u^2 - \sqrt{u+8}$, $u = \frac{v^2+6}{7}$ and $v = \frac{1}{(x-4)^3}$. [4 Marks]

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$x = 5$$

$$v = 1$$

$$u = 1$$

$$\frac{dy}{dx} = \left(4u - \frac{1}{\sqrt{u+8}}\right) \left(\frac{2v}{7}\right) \cdot \left(\frac{-3}{(x-4)^4}\right)$$

$$\frac{dy}{dx}\bigg|_{x=5} = \left(4 - \frac{1}{3}\right) \left(\frac{2}{7}\right) (-3) = -\frac{23}{7}$$

5. Given $h(x) = \frac{[f(x)]^2 + 1}{g(x^3)}$, where $h'(1) = 2$, $f(1) = 3$, $f'(1) = 2$ and $g(1) = 2$, determine the value of $g'(1)$. [4 Marks]

$$h'(x) = \frac{2f(x) \cdot f'(x) \cdot g(x^3) - ([f(x)]^2 + 1)(g'(x^3) \cdot 3x^2)}{(g(x^3))^2}$$

$$2 = \frac{2(3)(2) \cdot (2) - (9 + 1)(g'(1) \cdot 3(1)^2)}{4}$$

$$8 = 24 - 30g'(1)$$

$$g'(1) = \frac{8}{15}$$

Thinking - [5 Marks]

1. If $f(x) = x^3 - 1$ and $g(x) = x^2 - x$, determine the acute angle, in degrees, between the tangents to these graphs at the point of intersection of $f(x)$ and $g(x)$. Round your answer to 2 decimal places. [5 Marks]

$$f(x) = g(x)$$

$$f'(x) = 3x^2 \quad \left\{ \begin{array}{l} g'(x) = 2x - 1 \\ g'(1) = 1 \end{array} \right.$$

$$x^3 - 1 = x^2 - x$$

$$f'(1) = 3$$

$$g'(1) = 1$$

$$x^3 - x^2 + x - 1 = 0$$

$$x^2(x-1) + x-1 = 0 \Rightarrow \tan^{-1}(3) = 71.56^\circ$$

$$(x-1)(x^2+1) = 0$$

$$\tan^{-1}(1) = 45^\circ$$

$$\text{p.o.i. @ } (1, 0)$$

$\therefore$ Angle between tangents is 26.56°

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K & U	Thinking	Comm.
/20	/5	/2

Knowledge and Understanding - [20 Marks]

Multiple Choice: Write the CAPITAL LETTER corresponding to the correct answer on the line provided.
[1 Mark Each - 5 Marks Total]

- If $y = x^2 f\left(\frac{1}{x^3}\right)$, then $\frac{dy}{dx}$ is B
 - $x^2 f'\left(\frac{1}{x^3}\right)$
 - $2x f\left(\frac{1}{x^3}\right) - \frac{3}{x^2} f'\left(\frac{1}{x^3}\right)$
 - $2x f\left(\frac{1}{x^3}\right) + x^2 f'\left(\frac{1}{x^3}\right)$
 - $2x f'\left(\frac{1}{x^3}\right)$
- If $y = 2x^4 - 3x^3 - 6x + 23$, then $\left.\frac{d^3y}{dx^3}\right|_{x=-1}$ equals B
 - 6
 - 66
 - 30
 - 78
- $\lim_{h \rightarrow 0} \frac{[(x+h)^4 + 2(x+h)^3] - [x^4 + 2x^3]}{h}$ equals C
 - $x^4 + 2x^3$
 - $4x^3 - 6x^2$
 - $4x^3 + 6x^2$
 - Cannot be determined
- The position function of an object is given by $s(t) = t^3 - \frac{3}{2}t^2$, where time, t , is in seconds and distance, s , is in metres. When is the acceleration of the object equal to zero? A
 - $\frac{1}{2}$ seconds
 - 2 seconds
 - 4 seconds
 - 6 seconds
- Which of the following is/are true? C
 - If $g(x) = 3f(5x)$, then $g'(x) = 3f'(5x)$.
 - If $f(x) = \pi - 7$, then $f'(x) = 0$.
 - If $f(x)$ is a degree 7 polynomial function then $f^{(4)}(x)$ is a cubic polynomial function.
 - If $f(x) = g(x)h(x)t(x)$, then $f'(x) = g'(x)h(x)t(x) + g(x)h'(x)t(x) + g(x)h(x)t'(x)$.
 - II only
 - II and IV only
 - II, III and IV only
 - All are true
- Differentiate the following with respect to x . Do Not simplify your final answers. If applicable, leave answers with negative exponents. [6 Marks]

a. $y = \frac{7x^2 - 3x^5 - 10\sqrt[3]{x^8}}{x^5 - 7}$ [3]

$y = 7x^{-3} - 3 - 10x^{-\frac{7}{3}}$

$y' = -21x^{-4} + \frac{70}{3}x^{-\frac{10}{3}}$

b. $y = \frac{2}{(x-4)^3} - \sqrt[4]{3x + (x-9)^7} + \frac{\pi}{3}$ [3]

$y' = -6(x-4)^{-4} - \frac{1}{4}(3x + (x-9)^7)^{-\frac{3}{4}}(3 + 7(x-9)^6)$

7. Differentiate the following with respect to x . Completely simplify your answers. If applicable, write all answers with positive exponents. [9 Marks]

a. $f(x) = (6x-13)^4 \sqrt{5-3x^3}$ [4]

$$f'(x) = 4(6x-13)^3 \cdot 6 \cdot \sqrt{5-3x^3} + (6x-13)^4 \cdot \frac{1}{2} (5-3x^3)^{-\frac{1}{2}} (-9x^2)$$

$$= -3(6x-13)^3 (66x^3 - 39x^2 - 80)$$

$$= \frac{-3(6x-13)^3 (66x^3 - 39x^2 - 80)}{2\sqrt{5-3x^3}}$$

$$=$$

b. $h(x) = \frac{(2x^2-1)^3}{(5-2x)^4}$ [5]

$$h'(x) = \frac{3(2x^2-1)^2 \cdot (4x)(5-2x)^4 - (2x^2-1)^3 (4(5-2x)^3)(-2)}{(5-2x)^8}$$

$$= \frac{4(2x^2-1)^2 (5-2x)^3 [3x(5-2x) + 2(2x^2-1)]}{(5-2x)^8}$$

$$= \frac{4(2x^2-1)^2 (15x - 6x^2 + 4x^2 - 2)}{(5-2x)^5}$$

$$= \frac{4(2x^2-1)^2 (-2x^2 + 15x - 2)}{(5-2x)^5} = \frac{-4(2x^2-1)^2 (2x^2 - 15x + 2)}{(5-2x)^5}$$

Thinking - [5 Marks]

1. Given $f(x) = ax^3 + bx^2 + cx + d$, determine the values of a, b, c and d if $f''(0) = 2$, there are horizontal tangents at $x = -2$ and $x = 1$ and the function has a y -intercept equal to $g^{(4)}(4)$ where

$$g(x) = \frac{x^5}{120} - 6x^3 + 3x^2 + 1. \text{ [5 marks]}$$

$$f'(x) = 3ax^2 + 2bx + c \quad f''(x) = 6ax + 2b$$

$$f'(-2) = 0$$

$$12a - 4b + c = 0 \quad (1)$$

$$f'(1) = 0$$

$$3a + 2b + c = 0 \quad (2)$$

from (1) & (2)

$$12a - 4 + c = 0 \quad (3)$$

$$3a + 2 + c = 0 \quad (4)$$

$$f''(0) = 2$$

$$2b = 2$$

$$b = 1$$

$$(3) - (4)$$

$$9a - 6 = 0$$

$$a = \frac{2}{3}$$

sub $a = \frac{2}{3}$ into (3)

$$8 - 4 + c = 0$$

$$c = -4$$

$$g(x) = \frac{x^5}{120} - 6x^3 + 3x^2 + 1$$

$$g'(x) = \frac{5x^4}{120} - 18x^2 + 6x$$

$$g''(x) = \frac{20x^3}{120} - 36x + 6$$

$$g'''(x) = \frac{60x^2}{120} - 36, \quad g^{(4)}(x) = x$$

*** 2 marks will be awarded in the Communication Category for proper mathematical form. ***

$$\therefore g^{(4)}(4) = 4 \Rightarrow d = 4$$

UNIT 2 ASSESSMENT OF LEARNING: EXPLORING DERIVATIVES – DAY 2

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Application	Thinking	Comm.
/20	/5	/2

Application - [20 Marks]

1. A toy rocket is launched such that the position function, $s(t)$, in metres, of the toy rocket above the ground is $s(t) = -4t^3 + 24t^2 + \frac{1}{2}$, where t is in seconds.

- a. Determine the velocity of the toy rocket when the acceleration is 24 m/s^2 . [2 Marks]

$$v(t) = -12t^2 + 48t$$

$$a(t) = -24t + 48$$

$$24 = -24t + 48$$

$$t = 1$$

$$v(t) = 36 \text{ m/s}$$

- b. Determine the position of the rocket when the velocity -60 m/s . [3 Marks]

$$v(t) = -60 \Rightarrow -12t^2 + 48t = -60$$

$$t^2 - 4t - 5 = 0$$

$$(t-5)(t+1) = 0$$

$$\therefore t = 5, -1$$

$$\uparrow \text{reject}$$

$$s(5) = -4(5)^3 + 24(5)^2 + \frac{1}{2}$$

$$s(5) = 100.5 \text{ m}$$

2. Determine the equation(s), in point-slope form, of the line(s) through the point $(0, 2)$ that are tangent to the curve $f(x) = -2x^2 + 3x$. [4 Marks]

$$f'(x) = -4x + 3$$

$$y - y_1 = m(x - x_1)$$

$$-2x^2 + 3x - 2 = (-4x + 3)(x)$$

$$-2x^2 + 3x - 2 = -4x^2 + 3x$$

$$2x^2 - 2 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\therefore y - 2 = -x$$

$$\&$$

$$y - 2 = 7x$$

4. Using Leibniz's Notation, determine $\frac{dy}{dx} \Big|_{x=2}$ given $y = u^2 - 2u + 1$, $u = \sqrt{v} - \frac{v^2}{9}$ and $v = 3\sqrt{x^2 + 2x + 1}$. [4 Marks]

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$\frac{dy}{dx} = (2u - 2) \left(\frac{1}{2\sqrt{v}} - \frac{2v}{9} \right) \left(\frac{3(2x+2)}{2\sqrt{x^2+2x+1}} \right)$$

$$\frac{dy}{dx} \Big|_{x=2} = (-14) \left(\frac{1}{6} - 2 \right) (3)$$

$$\frac{dy}{dx} \Big|_{x=2} = 77$$

$$x = 2$$

$$v = 9$$

$$u = -6$$

3. Determine the slope of the tangent to the curve $y = \frac{x^3}{x^2-6}$ when $x = 3$. [3 Marks]

$$y' = \frac{3x^2(x^2-6) - x^3(2x)}{(x^2-6)^2}$$

$$m_t|_{x=3} = \frac{27(3) - 27(6)}{9}$$

$$m_t|_{x=3} = -9$$

5. Given $h(x) = \frac{[g(x)]^3 - 1}{f(x^2)}$, where $h(1) = 2$, $f(1) = 3$, $f'(1) = 2$ and $g(1) = 2$, determine the value of $g'(1)$. [4 Marks]

$$h'(x) = \frac{3[g(x)]^2 \cdot g'(x) f(x^2) - ([g(x)]^3 - 1) f'(x^2) \cdot 2x}{(f(x^2))^2}$$

$$2 = \frac{3(2)^2 g'(1)(3) - (8-1)(2)(2)(1)}{9}$$

$$18 = 36g'(1) - 28$$

$$46 = 36g'(1)$$

$$\therefore g'(1) = \frac{23}{18}$$

Thinking - [5 Marks]

1. Determine the equation(s), in point slope form, of the normal to the curve of $f(x) = -x^3 + 2x^2 - 2$ where the tangent has a positive slope and it makes an angle of $\frac{\pi}{4}$ with the x -axis. [5 Marks]

$$f(x) = -3x^2 + 4x$$

tangent angle of $\frac{\pi}{4} \Rightarrow \text{slope} = 1$

$$\therefore m_{\perp} = -1$$

$$-3x^2 + 4x = 1$$

$$3x^2 - 4x + 1 = 0$$

$$(3x-1)(x-1) = 0$$

$$x = \frac{1}{3}, 1$$

$$\left(\frac{1}{3}, -\frac{49}{27}\right)$$

$$(1, -1)$$

$$y - y_1 = m_{\perp}(x - x_1)$$

$$y + \frac{49}{27} = -(x - \frac{1}{3})$$

&

$$y + 1 = -(x - 1)$$